

RealCleanVSR: Mitigating the Perception-Distortion trade-off in Real-World Super-Resolution

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Abstract. Real-world video super-resolution (RWVSR) restores high-resolution videos from low-resolution observations affected by unknown blur, noise, and compression. A central challenge is the perception-distortion trade-off: suppressing degradations can improve perceptual quality but may also remove structures and textures required for faithful reconstruction. Existing RWVSR pipelines commonly apply residual cleaning before video refinement and upsampling, causing the later stages to rely mainly on cleaned information. We propose RealCleanVSR, a three-stage framework that mitigates this limitation through Dyna-Mix Cleaning and Self-Reference Upsampling. Dyna-Mix Cleaning maintains two complementary information flows: cleaned frames that reduce degradation artifacts and raw frames that preserve distortion-oriented details. These flows are combined during video refinement to support propagation, alignment, and aggregation without discarding useful input information. Self-Reference Upsampling then uses both flows as references and explores their complementary spatial cues in the high-resolution domain, where fine textures can be reconstructed more effectively. Experiments on real-world video super-resolution benchmarks show that RealCleanVSR surpasses RealBasicVSR in both PSNR and NIQE, demonstrating that it mitigates the perception-distortion trade-off. The proposed modules also transfer to real-world image super-resolution, where their integration improves the Real-ESRGAN baseline and demonstrates the broader applicability of the three-stage design across related real-world low-level computer vision and restoration tasks.

1 Introduction

Reconstructing a high-resolution (HR) video from its low-resolution (LR) counterpart is the objective of an active research field known as Video Super-Resolution (VSR) [12, 16, 18, 27]. Given the difficulty in obtaining aligned LR and HR data, Video Super-Resolution approaches choose to assume an ideal *bicubic downsampling kernel* and use a pixel-wise loss function [28, 35] to accurately reconstruct the video, correctness is judged using distortion metrics. When applied to Real-World Video Super-Resolution (RWVSR), the distortion-oriented methods [16, 45, 48] fail severely as the downsampling kernel becomes more complicated with the addition of unknown blur, noise, and compression. Conse-

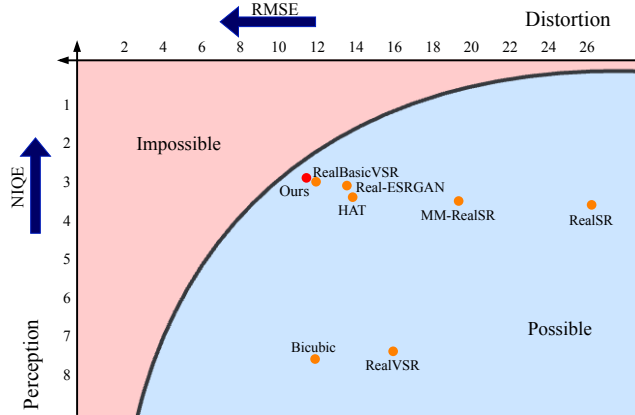


Fig. 1: Performance Gain on Perception-Distortion plane. We demonstrate the boundary on Perception-Distortion plane [11]. Our RealCleanVSR model outperforms the baselines in terms of both perception and distortion measures. Our model is represented by the red dot.

quently, this results in a continuous advancement in perception-oriented algorithms [42, 47], the goal of which is to fit the evaluation of human opinion.

A recent study, RealBasicVSR by Chan *et al.* [4] studies the effect of in-the-wild degradation carefully, then utilises a straightforward Residual Cleaning module prior to the set of refinement steps: Propagation, Alignment, Aggregation, and Residual Upsampling. The objective of Cleaning process is to estimate and subtract the degradation kernel from the input LR frames, converting the domain of these frames to the general *area downsampling* LR. However, in the trade-off between enhancing details and suppressing artifacts, the simple design of Residual Cleaning and Residual Upsampling in RealBasicVSR leans toward cartoon-style smoothness perception rather than fine detailedness distortion. It becomes imperative to devise more refined techniques that can effectively strike a balance, pushing the boundaries of Perception-Distortion further.

In this work, we design RealCleanVSR by developing Dyna-Mix Cleaning and Self-Reference Upsampling, two new modules that make it possible to investigate information in an efficient manner regarding both perception and distortion:

1. The proposed Dyna-Mix Cleaning alters the assumption of multi-oriented learning in previous Residual Cleaning methods [6, 43]. The Cleaning process now divides and then combines different feature flows to form a two-branch approach, where we obtain both cleaned frames with enhanced perception and raw frames with basic distortion. Each frame type has its unique benefits and can be combined to extract finer information, resulting in improved refinement in complicated regions, devoid of issues such as artifact propagation, misalignment, or false aggregation.

2. Residual Upsampling [23, 37] is the primary technique used for image-based upsampling, allowing the network architecture to concentrate entirely on learning the high-frequency residue. When incorporating Cleaning techniques into the design process, the decision between upscaling cleaned frames or utilizing raw frames can pose a challenge, given the absence of each visual reconstruction aspect. Additionally, the Video Refinement stages concentrate on LR space, neglecting to analyse the resampled HR space with rich spatial textures. Utilizing both the cleaned frames and raw frames obtained from the Cleaning stage, we introduce Self-Reference Upsampling, a method that concurrently achieves video resampling to the desired size while exploring HR expressiveness, leveraging both perception and distortion.

The two aforementioned components are novel and further discussion can be found in the respective component section. With its more effective designs, RealCleanVSR outperforms the latest state-of-the-art RWVSR RealBasicVSR in terms of perception and distortion. In particular, when compared to RealBasicVSR, a gain of 0.33 dB in PSNR and 0.03 in NIQE is obtained. Moreover, our two components are versatile and can easily adjust to image models. Hence, we present a novel adaptation of the famous Real-World Image Super-Resolution (RWISR) model Real-ESRGAN [40] by generalizing a three-stage approach from RWVSR, resulting in a notable improvement of 0.19 in NIQE and 0.42 dB in PSNR.

2 Related Work

Distortion-oriented Super-Resolution. As a pioneering work, Dong *et al.* [10] introduced SRCNN as an innovative approach to Super-Resolution (SR) using Convolutional Neural Networks (CNNs). The objective is to learn the transformation from LR to HR images with the use of distortion-oriented loss. Although trained on synthetic datasets, the reconstructed output yields unsatisfactory performance on both synthetic and real-world test sets. After SRCNN, several methodologies emphasizing model architecture [9, 16, 45, 48] were developed. For instance, Kim *et al.* [21] proposed a very deep CNN and residual learning, Lai *et al.* [24] introduced a Laplacian pyramid design on high-frequency residual learning, and Harris *et al.* [15] utilised deep back projection to learn resolution dependencies. In recent years, transformer networks [2, 30] have also been used as a new approach to the field of SR research. These advancements have led to a consistent enhancement in the visual quality of the SR outputs, facilitating the clear distinction between the synthetic reconstructed outputs and real-world counterparts. On synthetic test sets, reconstructed images tend to be blurry due to the pixel-wise approximation of distortion-oriented loss. On real-world test sets, reconstructed images suffer from broken image structure and strange HR detail because distortion-oriented training could not solve the problem of real-world perceptual generalization.

Perception-oriented Super-Resolution. To solve the problem of nonrealistic reconstruction, Ledig *et al.* [25] developed the SRGAN that employ adversarial loss [20] with previous distortion-oriented loss, leading to a new trend of perception-oriented methods [3, 39, 50]. As an enhanced version of SRGAN, ESRGAN [41] removed batch normalization and trained a deeper network with dense connection. To achieve both criteria of visual quality, ESRGAN trained a perception-oriented model and a distortion-oriented model, then performed parameter interpolation. These works are still limited in the statistics of the synthetic dataset and face reconstruction problems such as unrealistic details or unnatural structures on real-world test sets. Carefully analysing real-world degradation, Real-ESRGAN [40] proposed a second-order degradation model to create degraded training sets, intending to simulate the natural degradation model that is highly related to perception. Still, the reconstructed output cannot balance the missing distortion and create blended edges between non-semantic related objects.

3 Methodology

Our Real-World Clean Video Super-Resolution (RealCleanVSR) comprises two effective modules aimed at enhancing the processes of Cleaning and Upsampling. As depicted in Fig. 2, Dyna-Mix Cleaning is utilised to obtain distortion-rich and perception-rich features from the input video. These features are then propagated by the Propagation stage, where Alignment is accomplished as the subsequent step in the Video Super-Resolution (VSR) approach. Following the Alignment process, the next stage is Aggregation, which aims to generate video textures using convolution. Finally, Self-Reference Upsampling is applied at the end to upsample and explore HR space, resulting in high-quality super-resolved videos. The details and formulations of Dyna-Mix Cleaning and Self-Reference Upsampling are explained in the following sections.

3.1 Dyna-Mix Cleaning

Many Real-World Super-Resolution algorithms [4, 43] choose to adopt a pre-processing stage that is generally known as Residual Cleaning to achieve quality middle preprocessed frames (**cleaned frames**) or preprocessed features (cleaned features). Nevertheless, it has been observed that the assumption of Cleaning is not applicable to the *various degradation kernels* present in real-world scenarios [43]. This leads to the problem that while the Cleaning module effectively reduces artifacts in video frames, it also removes valuable textures within each image. Consequently, the preprocessed outputs align effectively with the perception expected in natural images but fail to work as a Super-Resolution technique due to the problem of distortion.

While perception and distortion are part of the trade-off presented in rate-distortion theory [7], previous works [1, 8, 11] indicate that the potential for devising an algorithm that excels in terms of both distortion and perception relies

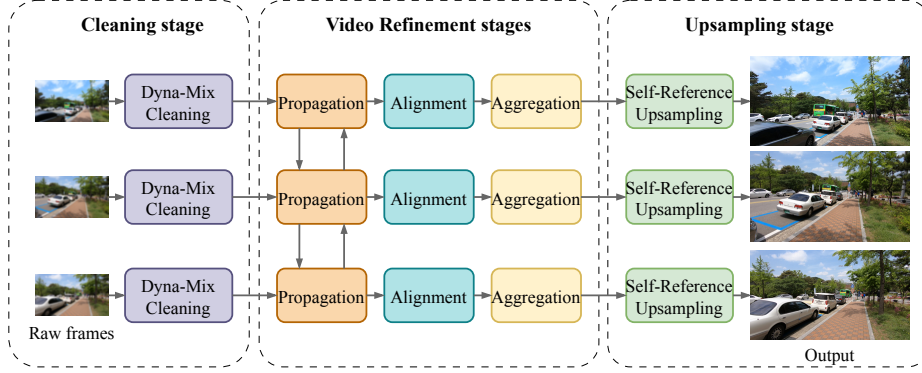


Fig. 2: An Overview of RealCleanVSR. RealCleanVSR comprises fundamental Video Refinement stages along with two novel, straightforward yet efficient modules for Cleaning and Upsampling processes. We introduce Dyna-Mix Cleaning to achieve mixed features between distortion and perception. In the final phase of the pipeline, Self-Reference Upsampling is proposed to resample and explore HR details.

on the integration of information from both aspects. Therefore, we use a multi-branch network [44] approach to design Dyna-Mix Cleaning where perception and distortion are explored with different sets of parameters.

In the first step of Dyna-Mix Cleaning, a residual block and dynamic refinement strategy [46] are employed to form the Dynamic Clean submodule. While the design of the residual block in Dynamic Clean is simple, we observe that the dynamic refinement strategy is robust and capable of generating high-quality perceptual cleaned frames for subsequent use. To extract perception-oriented features, cleaned frames are then proceeded to the cleaned feature encoder. Similarly, the raw feature encoder extracts distortion-oriented features from **raw frames** (initial input frames), forming the second branch of information retrieval. Mixed features are finally obtained by concatenating cleaned and raw features on a channel-wise basis, ready to be spatiotemporally explored by Video Refinement stages. Compared to existing approaches that rely on a single feature flow, Dyna-Mix Cleaning extracts more expressive detail from the entire sequence while pushing the Perception-Distortion bound. The graphical representation is depicted in Fig. 3(a).

The Dyna-Mix Cleaning module comprises three submodules, Dynamic Clean (*DC*), Raw Encoder (*RE*), and Cleaned Encoder (*CE*). Let x_t be the frame at time t of the input sequence, we obtain the cleaned frame $\hat{x}_t$ by iterating θ times:

$$\begin{cases} \hat{x}_t^0 = x_t, \\ \hat{x}_t^{j+1} = DC(\hat{x}_t^j), & \text{if } \text{mean}(|\hat{x}_t^j - \hat{x}_t^{j-1}|) \geq 1.0, \\ \hat{x}_t = \hat{x}_t^j, & \text{otherwise.} \end{cases} \quad (1)$$

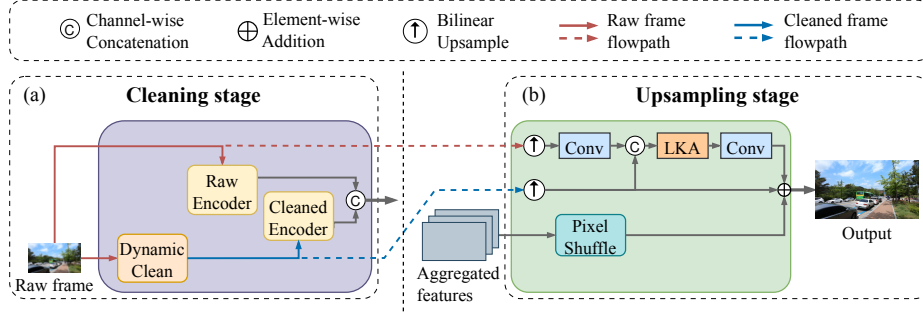


Fig. 3: Our Dyna-Mix Cleaning and Self-Reference Upsampling. (a) Dyna-Mix Cleaning module in RealCleanVSR generates cleaned frames by Dynamic Clean component, followed by obtaining mixed features through feature extraction from both distortion-rich raw frames and perception-rich cleaned frames. (b) Self-Reference Upsampling module facilitates HR feature exploration. By integrating attention, each pixel can directly contribute to the overall distortion and perception measurement.

As the output of Dyna-Mix Cleaning and the input to Video Refinement stages, the mixed feature f_i is obtained by the formula:

$$f_i = \text{Concat}(RE(x_i), CE(\hat{x}_i)). \quad (2)$$

Discussion. Both Blau *et al.* [1] and Freirich *et al.* [11] demonstrate that achieving the highest level of perception, which involves minimizing divergence from natural image statistics, is comparatively more attainable than reaching the optimal value of distortion. Therefore, the primary objective of our Dyna-Mix Cleaning approach is to enhance distortion while simultaneously preserving perception. This is crucial because even a slight increase in distortion for an algorithm that already excels in this aspect can notably deteriorate perception. We opt to utilise Dynamic Clean with a threshold of 1.0 and restrict the number of iterations $\theta = 3$ to minimize the loss of structural features in the cleaned frames; these configurations can be changed to obtain more perception or more distortion output. While combining features from the Raw Encoder, it is essential to recognise that although the textures extracted from raw frames are valuable, unforeseen degradation in real-world scenarios can result in corruption at the image level. These corruptions can create the expansion of artifact regions through the process of Propagation. Therefore, the number of features encoded from the raw frames should be set to less than the number of features encoded from the cleaned frames.

3.2 Self-Reference Upsampling

Upsampling stage is crucial in Super-Resolution since it significantly impacts the network’s learning process and its ability to achieve optimal performance.

In the following discussion, an analysis will be conducted on the advantages and disadvantages of each technique used in synthetic Upsampling. Additionally, we will introduce our novel Self-Reference module that effectively leverages both cleaned and raw frames to accomplish the real-world Upsampling process under the Perception-Distortion trade-off.

The synthetic Upsampling stage can follow either the approach of learning feature-based upsample [19, 40, 49] or learning residue-based upsample [4, 43]. The learning feature-based upsample, commonly referred to as Feature Upsampling [38], operates by performing image resampling methods such as bilinear, nearest neighbor, or pixel shuffle [36] at the feature level to attain the desired image size. The residual-based or Residual Upsampling strategy also employs a low-pass image resampling method but on image level. In this approach, the architecture is tasked with learning only the residue for the use of pixel-wise addition that create the final output frames.

- **Feature Upsampling.** Most methods [32, 40, 41, 47] generally do not perform Residual Upsampling, they rely on the architecture with many convolution blocks to refine and upsample the frames, offering flexibility in approaching the statistic of natural image. However, this approach is generally limited in distortion preservation, as image resampling methods are used on the feature level. During reconstructing context, these features are suppressed by convolution with some minor transitions in edges, hindering the ability to restore the structure of the image. It is important to note that the absence of distortion does not necessarily indicate good perceptual quality; therefore, the Upsampling process on features may fail in both aspects.
- **Residual Upsampling.** In the Cleaning stage, either cleaned or raw frames are utilised as the base for image resampling. However, the differences between these choices are insignificant, as both have their drawbacks. Raw frames are generally affected by the diversity of real-world artifacts, when subjected the Upsampling stage, the low-pass nature of image resampling filters further apply another layer of artifact enlargement to the frames. In the trade-off essence, resampled cleaned frames will serve better in visual perspective quality, while enduring the risk of losing image structural details compared to those which use raw frames.

Self-Reference Upsampling. Inspired by the recognition of Neural style transfer [44] and Reference-based Super-Resolution [26] as techniques capable of addressing the boundary between Perception-Distortion, our goal is to explore the potential of leveraging the double image approach. While our study focuses on standalone video frames without any style images or reference components, the HR components are restricted in the resampled cleaned and raw frames. Given the complex degradation in HR raw frames, we utilise HR cleaned frames as the base for upsampling. By incorporating an attention mechanism, our Self-Reference Upsampling is designed to transfer distortion-oriented details from HR raw frames to HR cleaned frames and reduce the trade-off behavior. The graphical representation is shown in Fig. 3(b).

Given the raw frames x_t and the cleaned frames $\hat{x}_t$, we obtained the upsample version u_t and $\hat{u}_t$ accordingly using the bilinear upsampling (*Bilinear*):

$$u_t = \text{Bilinear}(x_t) \quad (3)$$

$$\hat{u}_t = \text{Bilinear}(\hat{x}_t) \quad (4)$$

With *Conv* is the point-wise convolution layer, *LKA* is the Large-Kernel Attention module [14] and *PixShuf* is the pixel-shuffling technique [36] to upsample aggregated features ag_t , the final output video frame y_t is then achieved by the formula:

$$y_t = \text{PixShuf}(ag_t) + \text{Conv}(\text{LKA}(\text{Concat}(\hat{u}_t, \text{Conv}(u_t)))) + \hat{u}_t \quad (5)$$

Discussion. By comparing the orientation of cleaned and raw frames, we observe that the intuition of Sec. 3.1 is relevant in the context of HR space. Given that CNNs have restricted receptive fields that focus on local information, we have integrated an attention-based module. This module allows us to assign weights to distinct pixels, considering their contributions to the overall perception. As a result, we gain greater flexibility in incorporating global information into our processing. Furthermore, by focusing solely on the residue, the network can concentrate on learning subtle variations to effectively transfer information from raw frames to the enhanced version of cleaned frames. This approach minimises the occurrence of artifact growth, thus preventing potential degradation of visual quality.

4 Experiments

4.1 Experimental Settings

Training Settings. We use the synthetic REDS training dataset [34] and apply the second-order degradation model [40] to artificially create our real-world training dataset. In the second-order degradation model, random blur, resizing, noise, and JPEG compression are applied as image-based degradation. For video-based degradation, we incorporate video compression, which is a common technique to reduce video size. The training LR frames are extracted with a patch size of 64×64 in the spatial dimension, while a stochastic degradation strategy [4] is applied with a temporal length of 20 frames in the temporal dimension. We use Adam optimizer [22] with fixed learning rates for optimization. The training is divided into two phases: distortion-oriented phase and perception-oriented phase. We conducted experiments using a system equipped with an Nvidia A100 GPU with a memory capacity of 40GB. The information regarding model architecture is available in the supplementary material.

Distortion-oriented phase. As the distortion is harder to achieve, we pre-train RealCleanVSR using the pixel-wise loss of L1 for both output and Dynamic

Clean module. The pre-training is performed using a batch size of 8 and a learning rate of 10^{-4} with 400K iterations. Utilising distortion-oriented pre-training with pixel-wise L1 loss enables our model to address real-world degradation and prevent unwanted local optima.

Perception-oriented phase. As GAN-based training has been proved by Blau *et al.* [1] to approach the Perception-Distortion boundary, we further fine-tune the second phase of perception-oriented. We use RealCleanVSR as the generator and U-Net discriminator with spectral normalization (SN) [40] as the discriminator. In the perception-oriented phase, we fix the parameter of the Dynamic Clean module and refine the generator by incorporating both perceptual loss [20] and adversarial loss [13]. The batch size is decreased to 4. The learning rates of the generator and discriminator are configured at 5×10^{-5} and 10^{-4} , respectively. The number of iterations for the second phase is 150K.

Quantitative Metric. When evaluating visual restoration techniques, employing both non-reference and fully reference measurements is crucial. This approach ensures a reliable method of comparison, considering both perceptual quality and distortion. Therefore, we employ the widely used nonreference metric NIQE [31] to evaluate perception. Additionally, we utilise PSNR and SSIM [17] as full reference metrics to assess distortion.

4.2 Comparison to State-of-the-Arts

We compare our RealCleanVSR with five state-of-the-art methods, including three image models: Real-ESRGAN [40], MM-RealSR [32], HAT [5] and two video models: RealVSR [47], RealBasicVSR [4], as listed in Tab. 1.

For testing purposes, we use the REDS [34] test set, UDM10 [48] along with Vid4 [29]. All models are evaluated with $4\times$ upsample configurations.

Tab. 1 demonstrates RealCleanVSR’s superior performance across all three datasets. Specifically, our method surpasses the recent RWVSR method Real-BasicVSR [4] by achieving higher improvements of 0.33 dB in PSNR and 0.03 in NIQE on REDS test set, despite the trade-off characteristics of Perception-Distortion. showcases remarkable improvements, achieving a substantial enhancement of 0.85 on the NIQE measure compared to the state-of-the-art RWSR method HAT [5], as depicted in Tab. 1 on the UDM10 dataset. On VID4, we further achieve the best perception score with 0.01 NIQE better than Real-ESRGAN while securing the second-best PSNR. In the realm of RWSR, these advancements are considered noteworthy.

In addition, we calculate the Root Mean Square Error (RMSE) based on the Peak Signal-to-Noise Ratio (PSNR) score. The RMSE is used to reverse the direction of the *Distortion axis* towards zero value, where a lower value indicates better performance. We then visualise the Quadratic Perception-Distortion plane [11] using the NIQE and RMSE metrics on the REDS test dataset in Fig. 1. This

Table 1: Quantitative comparison (NIQE/PSNR/SSIM). All metrics are calculated on RGB-channel except NIQE [31] (Y-channel). **Red** and **blue** colors represent the best and second-best performance, respectively.

		Real-ESRGAN [40]	RealVSR [47]	MM-RealSR [33]	RealBasicVSR [4]	HAT [5]	RealCleanVSR
Source		CVPRW 2021	ICCV 2021	ECCV 2022	CVPR 2022	CVPR 2023	Ours
REDS [34]	NIQE ↓	3.0464	7.2672	3.3737	2.8467	3.3164	2.8190
	PSNR ↑	25.54	24.11	22.40	26.59	25.32	26.92
	SSIM ↑	0.7288	0.6555	0.6596	0.7432	0.7145	0.7533
UDM10 [48]	NIQE ↓	4.4727	8.0974	4.6126	4.0589	4.9999	4.1457
	PSNR ↑	23.94	23.87	22.78	24.09	24.54	24.18
	SSIM ↑	0.7648	0.7501	0.7406	0.7543	0.7692	0.7543
VID4 [29]	NIQE ↓	3.0830	7.8093	3.1347	3.1142	3.5477	3.0723
	PSNR ↑	18.80	19.92	17.73	19.10	19.05	19.11
	SSIM ↑	0.4676	0.4726	0.4466	0.4675	0.4745	0.4660

analysis includes a total of eight algorithms by adding a foundation method of RealSR [19] and bicubic upsampler.

In addition to quantitative measurements, we also provide a qualitative comparison for human perception assessment. The incorporation of the new Cleaning and Upsampling stage in RealCleanVSR enables effective alignment and combination of information in the Video Refinement stages, mitigating issues such as artifact growth or detail suppression. Consequently, it produces significantly more authentic details in intricate areas, enhancing the assessment of visual quality, as shown in Fig. 4.

5 Ablation Studies

5.1 Components analysis

To prove the effectiveness of our contributions, we retain the basic Video Refinement stages in Fig. 2, gradually applying previous methods and our proposed components to the two stages of Cleaning and Upsampling. The most noticeable quantitative gap, with a difference of 0.86 dB in distortion measure (PSNR), is observed in Tab. 2, demonstrating significant improvement across all metric results with the inclusion of our components.

Temporal Study. We thoroughly evaluate our idea based on creating temporal profiles using the method in Fig. 5(a). In the Ground Truth space, the temporal progression of natural video showcases seamless transitions, densely packed with intricate details throughout the pixel transitions. In order to achieve satisfactory reconstructed quality, a video reconstruction algorithm must build a curve transition profile (based on perception) with rich transition details (based on distortion).



Fig. 4: Qualitative Comparison. RealCleanVSR can efficiently analyse frame structures. It produces significantly more detailed information than previous works. Our model is the only algorithm that effectively reconstructs the building’s structure and the text "ZABITA 153" that resemble natural images.

Table 2: Ablation studies of previous methods and our proposed components. Our new modules lead to significant increases in both Perception and Distortion measurements compared to the previous methods.

Model	Cleaning	Upsampling	NIQE ($\downarrow$)	PSNR ($\uparrow$)
Baseline [4]	Residual	Feature	3.0607	26.06
	Residual	Residual	3.0689	26.50
	Dyna-Mix	Feature	2.7683	26.46
	Dyna-Mix	Residual	2.7759	26.66
Ours	Dyna-Mix	Self-Reference	2.8190	26.92

In Fig. 5(b), we present a comparison of the temporal profiles of the previous approaches with our novel components. Changing from Residual Cleaning to Dyna-Mix Cleaning enhances the network’s capacity to extract additional distortion information while preserving seamless transitions. Similarly, the switch from Residual Upsampling to Self-Reference Upsampling in RealCleanVSR effectively addresses aliasing issues, resulting in a more realistic transition curve.

Spatial Study. On spatial level, we further provide some qualitative comparisons to understand the contributions of the proposed Cleaning and Upsampling modules. As shown in the three examples of Fig. 6, in previous Cleaning strategy, the counterparts remove the edge texture and apply a smooth filter to the output. The Dyna-Mix Cleaning contribution initially focuses on the LR space to recover structural features, while the Self-Reference Upsampling technique operates in the HR domain to split and reconstruct complex regional details.

5.2 Design on Real-World Image Super-Resolution

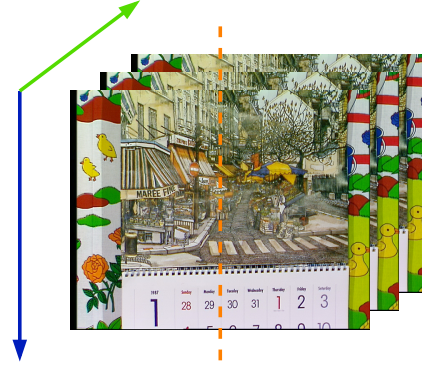
Utilizing our Dyna-Mix Cleaning and Self-Reference Upsampling image-level design, we further explore our proposal in the related task of RWISR using the architecture of Real-ESRGAN [40]. By integrating these components into the Real-ESRGAN baseline, our new Real-World Clean Super-Resolution GAN (Real-CSRGAN) model can achieve significant increases in both perception and distortion metrics. Quantitative improvement metrics on the REDS dataset are shown in Tab. 3. More information related to our image model are presented in the supplementary material.

Table 3: Ablation studies of our proposed components on RWISR. Our modules can easily adapt to image models and also lead to significant increases in both perception and distortion measurements.

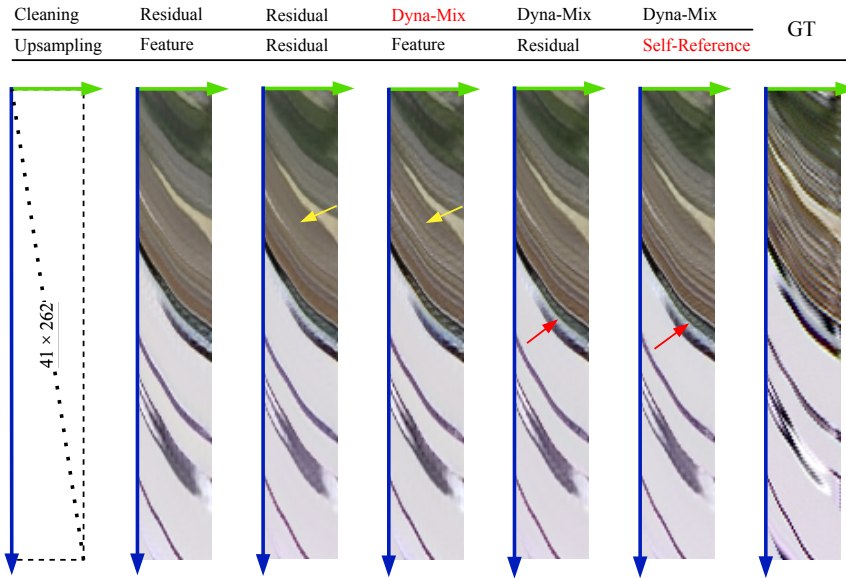
Model	NIQE (↓)	PSNR (↑)
Real-ESRGAN	3.0464	25.54
Real-CSRGAN (Ours)	2.8538	25.96

Based on the three-stage concept of RWVSR, we extend the RWISR approach to incorporate Cleaning, Refinement, and Upsampling stages. Given that a single image lacks the time dimension, compensating for degradation and textures involves making implicit predictions by stacking a large amount of Residual Blocks in the middle to create the Image Refinement stage.

We analyse the structure of Real-ESRGAN and divide it accordingly as shown in Fig. 7. In Real-ESRGAN, a convolution layer was previously utilised for feature extraction. However, as demonstrated in Sec. 3.1, this approach fails to address the multi-oriented objective of perception and distortion in RWISR. We



(a) Temporal profile



(b) Temporal profile comparison

Fig. 5: Temporal transition comparison. We compare the temporal transition of frames on the "Calendar" clip of the VID4 dataset [29]. The yellow arrow ($\uparrow$) indicates the transition from Residual Cleaning to Dyna-Mix Cleaning, while the red arrow ($\uparrow$) denotes the shift from previous Upsampling methods to our Self-Reference Upsampling. Our novel components substantially improve the reconstruction of temporal profiles.

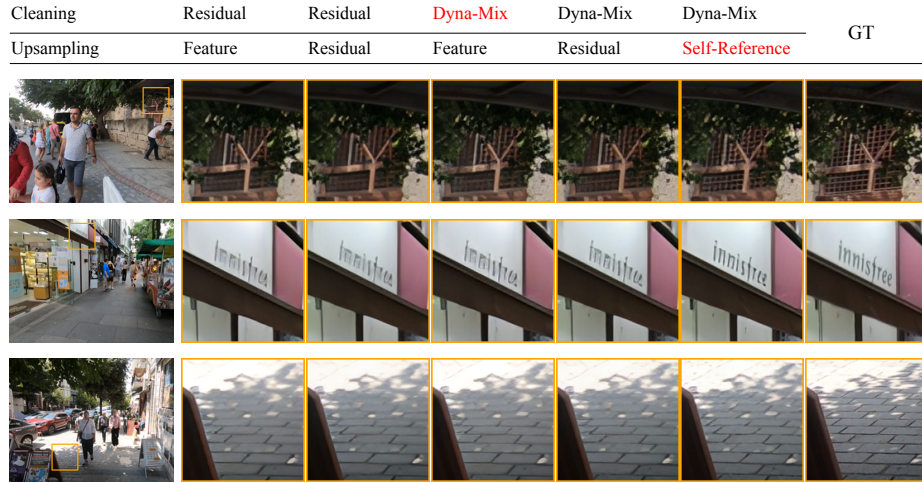


Fig. 6: Spatial comparison. Our two new components namely, Dyna-Mix Cleaning and Self-Reference Upsampling are able to explore both LR and HR spatial details to better restore video frames.

then substituted it with Dyna-Mix Cleaning, which is applied prior to the network. This modification allows us to obtain a preprocessed image and a combination of valuable features. In the Image Refinement stage, we maintain the high-level architecture design of numerous Residual-in-Residual Dense Blocks (RRDBs) [41] as they effectively explore deep and intricate structures for image restoration. In the Upsampling stage, we replace the inefficient Feature Upsampling with our Self-Reference Upsampling module. This transformation changes the Image Refinement process into a residual learning scheme, enabling us to execute residual scaling and ultimately element-wise addition to obtain the high-quality SR image.

6 Conclusion

In this study, we propose RealCleanVSR, introducing the novel Dyna-Mix Cleaning and Self-Reference Upsampling modules for RWVSR. Our RealCleanVSR model has demonstrated superior performance over the most advanced RWVR frameworks in terms of visual quality aspects, particularly perception and distortion, as evidenced by extensive experimental results. Both proposed components are adaptable and can seamlessly integrate into the image model. Additionally, with a robust design tailored to video and image models, we establish an efficient three-stage methodology that can serve as a foundational technique for real-world low-level vision tasks.

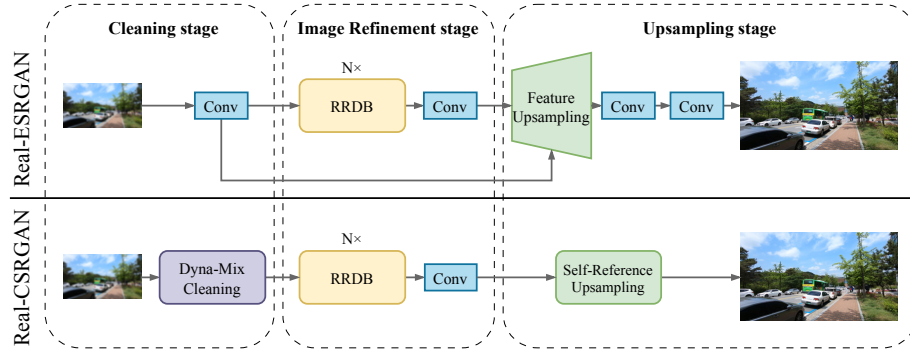


Fig. 7: Real-ESRGAN and new Real-CSRGAN architecture. In order to adapt the overall three-stage method Cleaning, Refinement, and Upsampling, we divide and substitute the Real-ESRGAN components with our modules. This simplifies the complex architecture of the RWISR method while establishing a comprehensive methodology for RWSR.

References

- Blau, Y., Michaeli, T.: The perception-distortion tradeoff. In: Proceedings of the IEEE conference on computer vision and pattern recognition. pp. 6228–6237 (2018) [4](#), [6](#), [9](#)
- Cao, J., Li, Y., Zhang, K., Van Gool, L.: Video super-resolution transformer. arXiv preprint arXiv:2106.06847 **2**(3), 7 (2021) [3](#)
- Chan, K.C., Wang, X., Xu, X., Gu, J., Loy, C.C.: Glean: Generative latent bank for large-factor image super-resolution. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 14245–14254 (2021) [4](#)
- Chan, K.C., Zhou, S., Xu, X., Loy, C.C.: Investigating tradeoffs in real-world video super-resolution. In: IEEE Conference on Computer Vision and Pattern Recognition (2022) [2](#), [4](#), [7](#), [8](#), [9](#), [10](#), [11](#)
- Chen, X., Wang, X., Zhou, J., Qiao, Y., Dong, C.: Activating more pixels in image super-resolution transformer. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 22367–22377 (2023) [9](#), [10](#)
- Chi, Y., Gu, J., Zhang, J., Yang, W., Tian, Y.: Egovsr: Towards high-quality egocentric video super-resolution (2023) [2](#)
- Cover, T.M., Thomas, J.A.: Elements of Information Theory (Wiley Series in Telecommunications and Signal Processing). Wiley-Interscience, USA (2006) [4](#)
- Deng, X., Yang, R., Xu, M., Dragotti, P.L.: Wavelet domain style transfer for an effective perception-distortion tradeoff in single image super-resolution. In: 2019 IEEE/CVF International Conference on Computer Vision (ICCV). pp. 3076–3085 (2019) [4](#)
- Dong, C., Loy, C.C., He, K., Tang, X.: Learning a deep convolutional network for image super-resolution. In: Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part IV 13. pp. 184–199. Springer (2014) [3](#)
- Dong, C., Loy, C.C., He, K., Tang, X.: Image super-resolution using deep convolutional networks. IEEE transactions on pattern analysis and machine intelligence **38**(2), 295–307 (2015) [3](#)

11. Freirich, D., Michaeli, T., Meir, R.: A theory of the distortion-perception tradeoff in wasserstein space. *Advances in Neural Information Processing Systems* **34**, 25661–25672 (2021) [2](#), [4](#), [6](#), [9](#)
12. Fuoli, D., Gu, S., Timofte, R.: Efficient video super-resolution through recurrent latent space propagation. In: *ICCV Workshops* (2019) [1](#)
13. Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial networks. *Communications of the ACM* **63**(11), 139–144 (2020) [9](#)
14. Guo, M.H., Lu, C.Z., Liu, Z.N., Cheng, M.M., Hu, S.M.: Visual attention network. *arXiv preprint arXiv:2202.09741* (2022) [8](#)
15. Haris, M., Shakhnarovich, G., Ukita, N.: Deep back-projection networks for super-resolution. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 1664–1673 (2018) [3](#)
16. Haris, M., Shakhnarovich, G., Ukita, N.: Recurrent back-projection network for video super-resolution. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 3897–3906 (2019) [1](#), [3](#)
17. Hore, A., Ziou, D.: Image quality metrics: Psnr vs. ssim. In: *2010 20th international conference on pattern recognition*. pp. 2366–2369. *IEEE* (2010) [9](#)
18. Isobe, T., Li, S., Jia, X., Yuan, S., Slabaugh, G., Xu, C., Li, Y.L., Wang, S., Tian, Q.: Video super-resolution with temporal group attention. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 8008–8017 (2020) [1](#)
19. Ji, X., Cao, Y., Tai, Y., Wang, C., Li, J., Huang, F.: Real-world super-resolution via kernel estimation and noise injection. In: *proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*. pp. 466–467 (2020) [7](#), [10](#)
20. Johnson, J., Alahi, A., Fei-Fei, L.: Perceptual losses for real-time style transfer and super-resolution (2016) [4](#), [9](#)
21. Kim, J., Lee, J.K., Lee, K.M.: Accurate image super-resolution using very deep convolutional networks. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 1646–1654 (2016) [3](#)
22. Kingma, D.P., Ba, J.: Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014) [8](#)
23. Konno, Y., Tanaka, M., Okutomi, M., Yanagawa, Y., Kinoshita, K., Kawade, M.: Depth map upsampling by self-guided residual interpolation. In: *2016 23rd International Conference on Pattern Recognition (ICPR)*. pp. 1394–1399. *IEEE* (2016) [3](#)
24. Lai, W.S., Huang, J.B., Ahuja, N., Yang, M.H.: Deep laplacian pyramid networks for fast and accurate super-resolution. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 624–632 (2017) [3](#)
25. Ledig, C., Theis, L., Huszár, F., Caballero, J., Cunningham, A., Acosta, A., Aitken, A., Tejani, A., Totz, J., Wang, Z., et al.: Photo-realistic single image super-resolution using a generative adversarial network. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 4681–4690 (2017) [4](#)
26. Lee, J., Lee, M., Cho, S., Lee, S.: Reference-based video super-resolution using multi-camera video triplets. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 17824–17833 (2022) [7](#)
27. Li, Y., Zhu, H., He, L., Wang, D., Shi, J., Wang, J.: Video super-resolution with regional focus for recurrent network. *Applied Sciences* **13**(1), 526 (2022) [1](#)
28. Lim, B., Son, S., Kim, H., Nah, S., Mu Lee, K.: Enhanced deep residual networks for single image super-resolution. In: *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*. pp. 136–144 (2017) [1](#)

29. Liu, C., Sun, D.: On bayesian adaptive video super resolution. *IEEE transactions on pattern analysis and machine intelligence* **36**(2), 346–360 (2013) [9](#), [10](#), [13](#)
30. Lu, Z., Li, J., Liu, H., Huang, C., Zhang, L., Zeng, T.: Transformer for single image super-resolution. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 457–466 (2022) [3](#)
31. Mittal, A., Soundararajan, R., Bovik, A.C.: Making a “completely blind” image quality analyzer. *IEEE Signal processing letters* **20**(3), 209–212 (2012) [9](#), [10](#)
32. Mou, C., Wu, Y., Wang, X., Dong, C., Zhang, J., Shan, Y.: Metric learning based interactive modulation for real-world super-resolution. In: *European Conference on Computer Vision*. pp. 723–740. Springer (2022) [7](#), [9](#)
33. Mou, C., Wu, Y., Wang, X., Dong, C., Zhang, J., Shan, Y.: Metric learning based interactive modulation for real-world super-resolution. In: *European Conference on Computer Vision*. pp. 723–740. Springer (2022) [10](#)
34. Nah, S., Baik, S., Hong, S., Moon, G., Son, S., Timofte, R., Mu Lee, K.: Ntire 2019 challenge on video deblurring and super-resolution: Dataset and study. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. pp. 0–0 (2019) [8](#), [9](#), [10](#)
35. Nah, S., Hyun Kim, T., Mu Lee, K.: Deep multi-scale convolutional neural network for dynamic scene deblurring. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 3883–3891 (2017) [1](#)
36. Qin, M., Mavromatis, S., Hu, L., Zhang, F., Liu, R., Sequeira, J., Du, Z.: Remote sensing single-image resolution improvement using a deep gradient-aware network with image-specific enhancement. *Remote Sensing* **12**(5), 758 (2020) [7](#), [8](#)
37. Tomar, N.K., Bagci, U., Jha, D.: Rupnet: residual upsampling network for real-time polyp segmentation. In: *Medical Imaging 2023: Computer-Aided Diagnosis*. vol. 12465, pp. 368–374. SPIE (2023) [3](#)
38. Wang, J., Chen, K., Xu, R., Liu, Z., Loy, C.C., Lin, D.: Carafe: Content-aware reassembly of features. In: *Proceedings of the IEEE/CVF international conference on computer vision*. pp. 3007–3016 (2019) [7](#)
39. Wang, X., Li, Y., Zhang, H., Shan, Y.: Towards real-world blind face restoration with generative facial prior. In: *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. pp. 9168–9178 (2021)
40. Wang, X., Xie, L., Dong, C., Shan, Y.: Real-esrgan: Training real-world blind super-resolution with pure synthetic data. In: *Proceedings of the IEEE/CVF international conference on computer vision*. pp. 1905–1914 (2021) [3](#), [4](#), [7](#), [8](#), [9](#), [10](#), [12](#)
41. Wang, X., Yu, K., Wu, S., Gu, J., Liu, Y., Dong, C., Qiao, Y., Change Loy, C.: Esrgan: Enhanced super-resolution generative adversarial networks. In: *Proceedings of the European conference on computer vision (ECCV) workshops*. pp. 0–0 (2018) [4](#), [7](#), [14](#)
42. Wu, Y., Wang, X., Li, G., Shan, Y.: Animesr: learning real-world super-resolution models for animation videos. *Advances in Neural Information Processing Systems* **35**, 11241–11252 (2022) [2](#)
43. Xie, L., Wang, X., Shi, S., Gu, J., Dong, C., Shan, Y.: Mitigating artifacts in real-world video super-resolution models. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 37, pp. 2956–2964 (2023) [2](#), [4](#), [7](#)
44. Xu, Y., Zhang, N., Li, L., Sang, G., Zhang, Y., Wang, Z., Wei, P.: Joint learning of super-resolution and perceptual image enhancement for single image. *IEEE Access* **9**, 48446–48461 (2021) [5](#), [7](#)

45. Xue, T., Chen, B., Wu, J., Wei, D., Freeman, W.T.: Video enhancement with task-oriented flow. *International Journal of Computer Vision* **127**, 1106–1125 (2019) [1](#), [3](#)
46. Yang, J., Bhalgat, Y., Chang, S., Porikli, F., Kwak, N.: Dynamic iterative refinement for efficient 3d hand pose estimation. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. pp. 1869–1879 (2022) [5](#)
47. YANG, X., Xiang, W., Zeng, H., Zhang, L.: Real-world video super-resolution: A benchmark dataset and a decomposition based learning scheme. *ICCV* (2021) [2](#), [7](#), [9](#), [10](#)
48. Yi, P., Wang, Z., Jiang, K., Jiang, J., Ma, J.: Progressive fusion video super-resolution network via exploiting non-local spatio-temporal correlations. In: *Proceedings of the IEEE/CVF international conference on computer vision*. pp. 3106–3115 (2019) [1](#), [3](#), [9](#), [10](#)
49. Zhang, K., Liang, J., Van Gool, L., Timofte, R.: Designing a practical degradation model for deep blind image super-resolution. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 4791–4800 (2021) [7](#)
50. Zhang, W., Liu, Y., Dong, C., Qiao, Y.: Ranksgan: Generative adversarial networks with ranker for image super-resolution. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 3096–3105 (2019) [4](#)